DATA SCIENCE

The German Tank Problem

Estimating your chances of winning the lottery with

Dorian Drost
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Statistical estimates can be fascinating, can't t
sampling a few instances from a population, yo
properties of that population such as the mean
variance. Likewise, under the right circumstanc
to estimate the **total size** of the population, as
you in this article.

I will use the example of drawing lottery ticket
many tickets there are in total, and hence calc
of winning. More formally, this means to estima
size given a discrete uniform distribution. We v
estimates and discuss their differences and we
addition, I will point you to some other use cas
can be used in.

Playing the lottery

I'm too anxious to ride one of those, but if it pleases you...Photo by O

Let's imagine I go to a state fair and buy some
lottery. As a data scientist, I want to know the
winning the main prize, of course. Let's assum
single ticket that wins the main prize. So, to es
likelihood of winning, I need to know the total
tickets N, as my chance of winning is 1/N then
tickets). But how can I estimate that N by just
tickets (which are, as I saw, all losers)?

I will make use of the fact, that the lottery tick
on them, and I assume, that these are consecu
numbers (which means, that I assume a discre
distribution). Say I have bought some tickets a
in order are [242,412,823,1429,1702]. What do I k
total number of tickets now? Well, obviously th
1702 tickets (as that is the highest number I ha
at gives me a first lower bound of the numb

how accurate is it for the actual number of tickets? Just because the highest number I have drawn is 1702, that doesn't mean that there are any numbers higher than that. It is very unlikely, that I caught the lottery ticket with the highest number in my sample.

However, we can make more out of the data. Let us think as follows: If we knew the middle number of all the tickets, we could easily derive the total number from that: number is *m*, then there are *m-1* tickets below number, and there are *m-1* tickets above that. number of tickets would be *(m-1) + (m-1) + 1*, (v the ticket of number m itself), which is equal t know that middle number m, but we can estim or the median of our sample. My sample above average of 922, which yields 2*922-1 = 1843. Th calculation the estimated number of tickets is

That was quite interesting so far, as just from a numbers, I was able to give an estimate of the tickets. However, you may wonder if that is the can get. Let me spoil you right away: It is not.

The method we used has some drawbacks. Let that to you with another example: Say we have [12,30,88], which leads us to 2*43–1 = 85. That formula suggests there are 85 tickets in total. ticket number 88 in our sample, so this can nc There is a general problem with this method: T can be lower than the highest number in the s case, the estimate has no meaning at all, as w that the highest number in the sample is a nat of the overall N.

A better approach: Using even spacing

These birds are quite evenly spaced on the power line, which is an important
Photo by Ridham Nagralawala on Unsplash

Okay, so what can we do? Let us think in a diff
The lottery tickets I bought have been samplec
the distribution that goes from 1 to unknown N
the highest number is number 1702, and I wonc
is this from being the highest ticket at all. In ot
the gap between 1702 and N? If I knew that gaı
calculate N from that. What do I know about th
Well, I have reason to assume that this gap is e
big as all the other gaps between two consecu
sample. The gap between the first and the sec
on average, be as big as the gap between the s
third ticket, and so on. There is no reason why
should be bigger or smaller than the others, ex
viation, of course. I sampled my lottery ticke
so they should be evenly spaced on the range

ticket numbers. On average, the numbers in the range of 0 to N would look like birds on a power line, all having the same gap between them.

That means I expect N-1702 to equal the average of all the other gaps. The other gaps are 242–0=242, 412–242=170, 823–412=411, 1429–823=606, 1702–1429=273, which gives the average 340. Hence I estimate N to be 1702+340=2042. In sh denoted by the following formula:

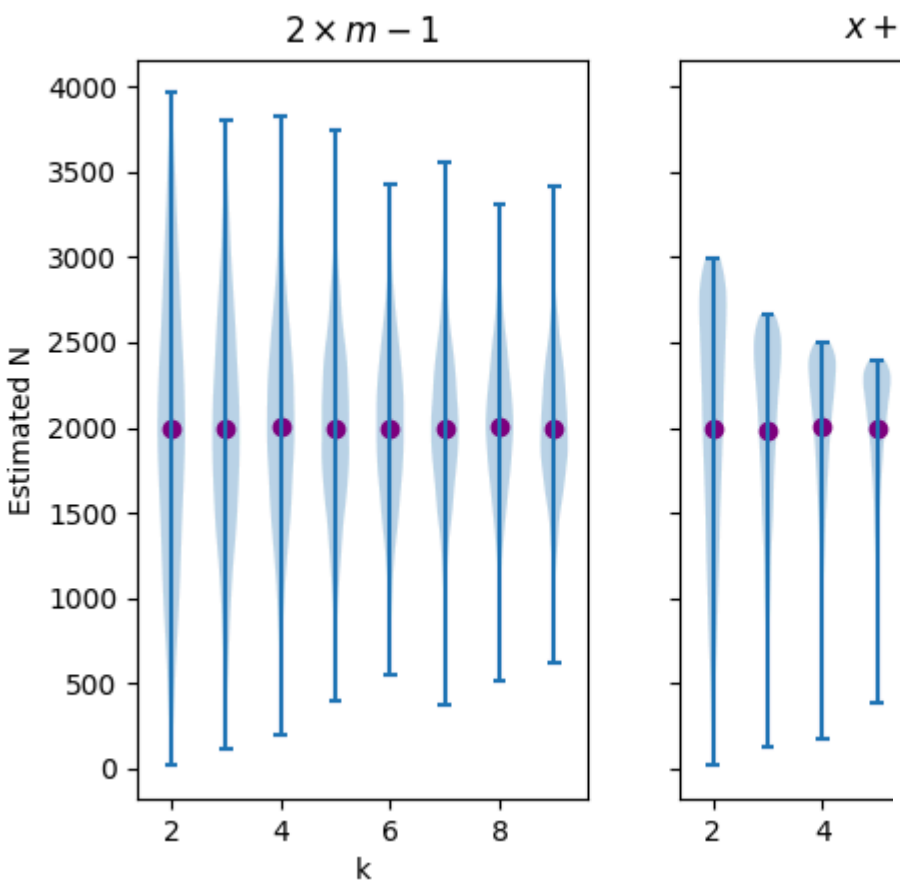
$$N = x + \frac{x - k}{k}$$

Here *x* is the biggest number observed (1702, ir is the number of samples (5, in our case). This form of calculating the average as we just did.

Let's do a simulation

We just saw two estimates of the total numbe First, we calculated *2*m-1*, which gave us 1843 the more sophisticated approach of *x + (x-k)/k* 2042. I wonder which estimation is more corre chances of winning the lottery 1/1843 or 1/2042

To show some properties of the estimates we j simulation. I drew samples of different sizes k distribution, where the highest number is 200(few hundred times each. Hence we would expe estimates also return 2000, at least on average outcome of the simulation:



Probability densities of the different estimates for varying k. Note that the gr
author.

What do we see here? On the x-axis, we see th
number of samples we take. For each k, we se
of the estimates based on a few hundred simu
formulas we just got to know. The dark point ir
value of the simulations each, which is always
of the k. That is a very interesting point: Both e
to the correct value if they are repeated an infi
times.

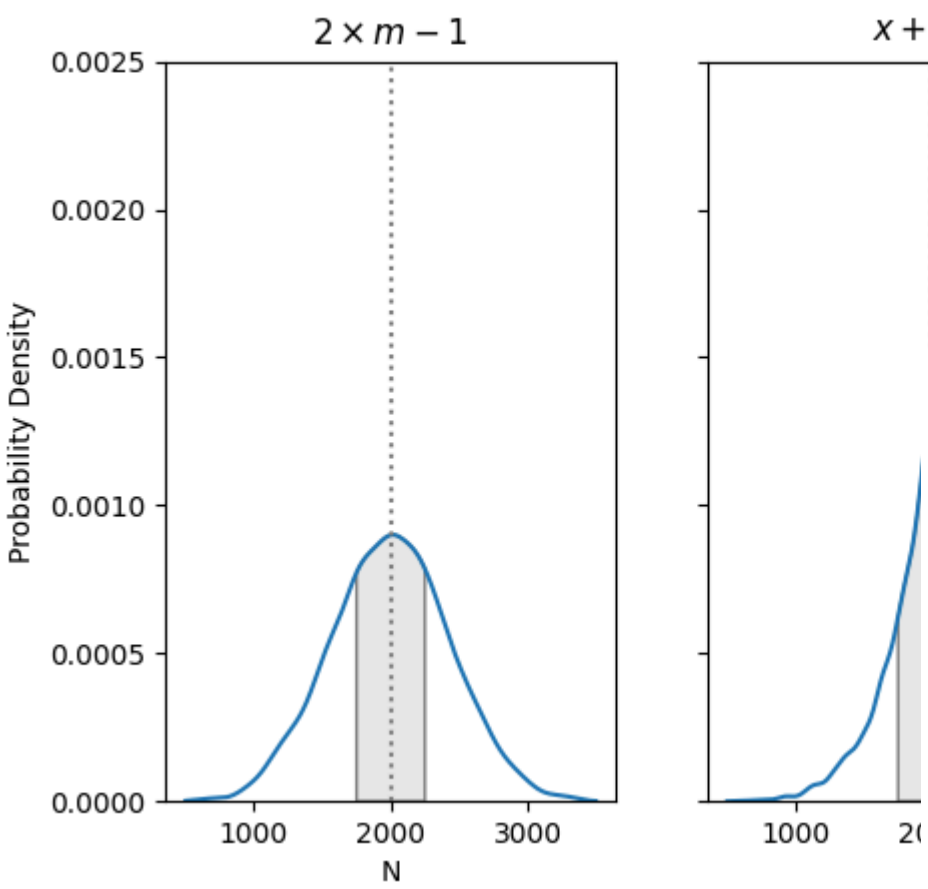
However, besides the common average, the dis
quite a lot. We see, that the formula *2*m-1* has
i.e. its estimates are far away from the real valu
for the other formula. The variance has a tende
with higher k though. This decrease does not a
rfectly, as this is just as simulation and is st

random influences. However, it is quite understandable and expected: The more samples I take, the more precise is my estimation. That is a very common property of statistical estimates.

We also see that the deviations are symmetrical, i.e. underestimating the real value is as likely as overestimating it. For the second approach, this symmetry does not hold. Most of the density is above the real mean, the larger outliers below. How does that come? Let computed that estimate. We took the biggest number in our sample and added the average gap size to that. The biggest number in our sample can only be as big as the number in total (the N that we want to estimate). We add the average gap size to N, but we can't get that with our estimate. In the other direction, the estimate can be very low. If we are unlucky, we could draw [1,2,3,4,5], in which case the biggest number in our sample is very far away from the actual N. That is why larger deviations are possible in underestimating the real value than overestimating it.

Which is better?

From what we just saw, which estimate is better? Both give the correct value on average. However, the estimate *k)/k* has lower variance, and that is a big advantage. That you are closer to the real value more often. I will demonstrate that to you. In the following, you see the density plots of the two estimates for a sample



Probability densities for the two estimates for k=5. The colored shape under t
from N=1750 to N=2250. Image by author.

I highlighted the point N=2000 (the real value f
line. First of all, we still see the symmetry that
before already. In the left plot, the density is d
symmetrically around N=2000, but in the right
to the right and has a longer tail to the left. No
at the grey area under the curves each. In both
from N=1750 to N=2250. However, in the left pl
accounts for 42% of the total area under the cu
right plot, it accounts for 73%. In other words,
have a chance of 42% that your estimate is **not**
than 250 points in either direction. In the right
is 73%. That means, you are much more likely t
the real value. However, you are more likely to
overestimate than underestimate.

I can tell you, that x*+ (x-k)/k* is the so-called **uniformly minimum variance unbiased estimator**, i.e. it is the estimator with the smallest variance. You won't find any estimate with lower variance, so this is the best you can use, in general.

Use cases

Make love, not war 💙 . Photo by Marco Xu on Unsp

We just saw how to estimate the total number
pool, if these elements are indicated by consec
Formally, this is a discrete uniform distribution
commonly known as the **German tank problem**
World War, the Allies used this approach to est
tanks the German forces had, just by using the
the tanks they had destroyed or captured so fa

We can now think of more examples where we
proach. Some are:

- You can estimate how many instances of a product have been produced if they are labeled with a running serial number.
- You can estimate the number of users or customers if you are able to sample some of their IDs.
- You can estimate how many students are (or have been) at your university if you sample students' mat numbers (given that the university has not numbers again after reaching the maximun

However, be aware that some requirements ne to use that approach. The most important one draw your samples randomly and independentl you ask your friends, who have all enrolled in t their matriculation numbers, they won't be eve whole range of matriculation numbers but will Likewise, if you buy articles with running numb you need to make sure, that this store got thes random fashion. If it was delivered with the pr 1000 to 1050, you don't draw randomly from th

Conclusion

We just saw different ways of estimating the to instances in a pool under discrete uniform dist both estimates give the same expected value i they differ in terms of their variance, with one the other. This is interesting because neither o wrong or right. Both are backed by reasonable considerations and estimate the real populatio frequentist statistical terms).

I now know that my chance of winning the state fair lottery is estimated to be 1/2042 = 0.041% (or 0.24% with the 5 tickets I bought). Maybe I should rather invest my money in cotton candy; that would be a save win.

References & Literature

Mathematical background on the estimates dis
article can be found here:

- Johnson, R. W. (1994). Estimating the size c
  Teaching Statistics, *16*(2), 50–52.

Also feel free to check out the Wikipedia articl
tank problem and related topics, which are qui

- https://en.wikipedia.org/wiki/German_tank_
- https://en.wikipedia.org/wiki/Discrete_unifc

This is the script to do the simulation and crea
in the article:

```
import numpy as np
import random
from scipy.stats import gaussian_kde
import matplotlib.pyplot as plt

if __name__ == "__main__":
    N = 2000
    n_simulations = 500

    estimate_1 = lambda sample: 2 * round(np.m
    estimate_2 = lambda sample: round(max(samp

    estimate_1_per_k, estimate_2_per_k = [],[]
    k_range = range(2,10)
    for k in k_range:
        diffs_1, diffs_2 = [],[]
```

```
    # sample without duplicates:
    samples = [random.sample(range(N), k) for _ in range(n_sim
    estimate_1_per_k.append([estimate_1(sample) for sample in
    estimate_2_per_k.append([estimate_2(sample) for sample in

fig,axs = plt.subplots(1,2, sharey=True, sharex=True)
axs[0].violinplot(estimate_1_per_k, positions=k_range, showext
axs[0].scatter(k_range, [np.mean(d) for d in estimate_1_per_k]
axs[1].violinplot(estimate_2_per_k, positi
axs[1].scatter(k_range, [np.mean(d) for d

axs[0].set_xlabel("k")
axs[1].set_xlabel("k")
axs[0].set_ylabel("Estimated N")
axs[0].set_title(r"$2times m-1$")
axs[1].set_title(r"$x+frac{x-k}{k}$")
plt.show()

plt.gcf().clf()
k = 5
xs = np.linspace(500,3500, 500)

fig, axs = plt.subplots(1,2, sharey=True)
density_1 = gaussian_kde(estimate_1_per_k[
axs[0].plot(xs, density_1(xs))
density_2 = gaussian_kde(estimate_2_per_k[
axs[1].plot(xs, density_2(xs))
axs[0].vlines(2000, ymin=0, ymax=0.003, co
axs[1].vlines(2000, ymin=0, ymax=0.003, co
axs[0].set_ylim(0,0.0025)

a,b = 1750, 2250
ix = np.linspace(a,b)
verts = [(a, 0), *zip(ix, density_1(ix)),
poly = plt.Polygon(verts, facecolor='0.9',
axs[0].add_patch(poly)
print("Integral for estimate 1: ", density

verts = [(a, 0), *zip(ix, density_2(ix)),
poly = plt.Polygon(verts, facecolor='0.9',
axs[1].add_patch(poly)
print("Integral for estimate 2: ", density
```

```
axs[0].set_ylabel("Probability Density")
axs[0].set_xlabel("N")
axs[1].set_xlabel("N")
axs[0].set_title(r"$2\times m-1$")
axs[1].set_title(r"$x+\frac{x-k}{k}$")

plt.show()
```

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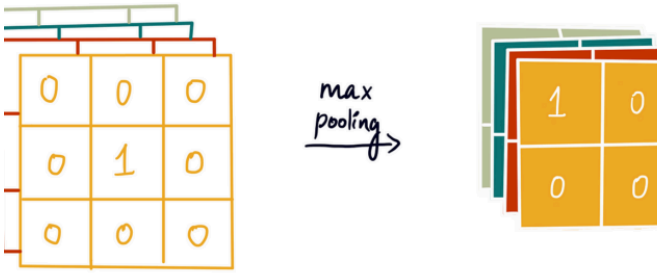

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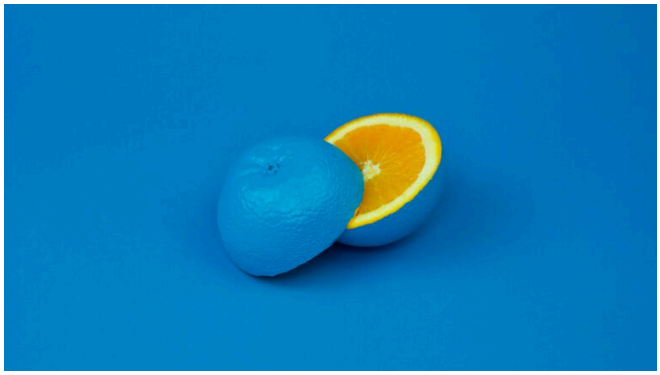

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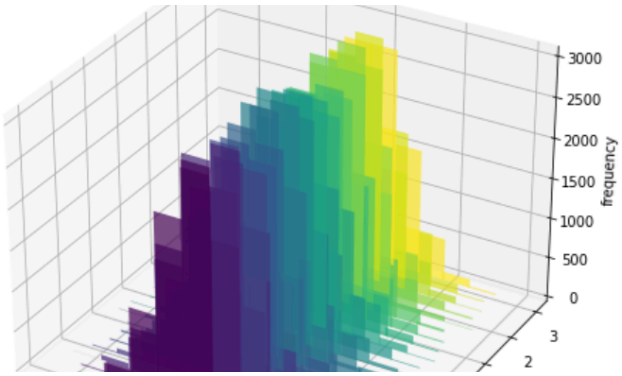

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